

Workshop on Advanced Sampling Methodologies

Advanced R Programming

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Workshop's agenda

- **Day 1:** Introduction to QGIS
- **Day 2:** Automatic update of census EA and national sampling frame
- **Day 3:** Introduction to R programming
- **Day 4:** Introduction to sampling design
- **Day 5: Advanced R programming**
- **Day 6:** Earth observation (EO) for sampling design
- **Day 7:** AI and machine learning for sample selection
- **Day 8:** Simulation and optimization of sampling strategies
- **Day 9:** R Shiny for sampling design evaluation
- **Day 10:** Wrap-up and workshop evaluation

Course materials

1. Today's course materials are on **Github**:
https://github.com/gianlucaboo/sampling_design_workshop
2. Click on **Day 5: Advanced R programming**
3. **Copy the url** and **paste it** on <https://download-directory.github.io>
4. Save the zipped directory on the workshop and unzip it

Today's agenda

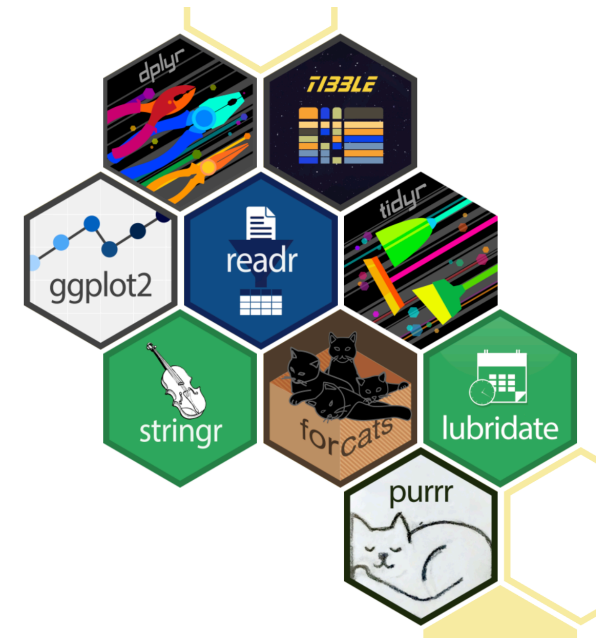
- The **tidyverse**
- Pipes (**|>** and **%>%**)
- Data manipulation with **dplyr**
- Data visualization with **ggplot2**
- Practice exercises

The tidyverse

- A collection of R packages for **data science**
- Common design philosophy and consistent syntax

Core packages:

- **dplyr**: data manipulation
- **ggplot2**: data visualization
- **tidyr**: data cleaning
- **readr**: data import
- **stringr**, **forcats**, **purrr**, etc.



Install once:

```
1 install.packages("tidyverse")
```

Load into session:

```
1 library(tidyverse)
```

Pipes

- Pipes make code **read left-to-right**
- **Combine functions** in a readable manner
- Avoids deeply **nested functions**
- Ctrl+Shift+M (Windows) or Cmd+Shift+M (Mac)

Two different **types of pipes** exists:

- Base R pipe: `|>`
- Magrittr pipe: `%>%`

Example: selecting the top 2 cars from mtcars with mpg over 20, sorted from highest to lowest mpg.

Without pipes:

```
1 library(tidyverse)
2 head(arrange(filter(mtcars, mpg > 20),
```

With pipes:

```
1 library(tidyverse)
2 mtcars |>
3   filter(mpg > 20) |>
4   arrange(desc(mpg)) |>
5   head(2)
```

 Try to run the code in your Script Editor.

Data manipulation with dplyr

- Part of the **tidyverse**
- Makes data manipulation **easy and readable**
- Works primarily with **data frames/tibbles**

Function	Purpose	Example
<code>filter()</code>	Subset rows based on conditions	<code>filter(mtcars, mpg > 20)</code>
<code>arrange()</code>	Sort rows	<code>arrange(mtcars, desc(mpg))</code>
<code>select()</code>	Pick columns	<code>select(mtcars, mpg, cyl)</code>
<code>mutate()</code>	Create or modify columns	<code>mutate(mtcars, kpl = mpg * 0.425)</code>
<code>summarise()</code>	Summarize data	<code>summarise(mtcars, avg_mpg = mean(mpg))</code>

dplyr::filter()

- **Purpose:** subset rows based on conditions
- **Syntax:** `filter(data, condition)`
- **Example:** filter cars above 20 mpg

```
1 library(tidyverse)
2 mtcars |>
3 filter(mpg > 20)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2

dplyr::arrange()

- **Purpose:** sort rows by column values
- **Syntax:** `arrange(data, column1, desc(column2))`
- **Example:** sort cars from highest to lowest mpg

```
1 library(tidyverse)
2 mtcars |>
3 arrange(desc(mpg))
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4

dplyr::select()

- **Purpose:** choose specific columns
- **Syntax:** `select(data, col1, col2, ...)`
- **Example:** returns only mpg, cyl, and hp columns

```
1 library(tidyverse)
2 mtcars |>
3 select(mpg, cyl, hp)
```

	mpg	cyl	hp
Mazda RX4	21.0	6	110
Mazda RX4 Wag	21.0	6	110
Datsun 710	22.8	4	93
Hornet 4 Drive	21.4	6	110
Hornet Sportabout	18.7	8	175
Valiant	18.1	6	105
Duster 360	14.3	8	245
Merc 240D	24.4	4	62
Merc 230	22.8	4	95
Merc 280	19.2	6	123
Merc 280C	17.8	6	123
Merc 450SE	16.4	8	180
Merc 450SL	17.3	8	180

dplyr::mutate()

- **Purpose:** add or modify columns
- **Syntax:** `mutate(data, new_col = expression)`
- **Example:** add a new column converting mpg to km/l

```
1 library(tidyverse)
2 mtcars |>
3 mutate(kpl = mpg * 0.425)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
kpl											
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
8.9250											
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
8.9250											
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
9.6900											
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
9.0950											
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
7.9475											
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
7.6925											

summarise() and group_by()

- **Purpose:** aggregate data
- **Syntax:** `data %>% group_by(group_col) %>% summarise(summary_col = mean(value))`
- **Example:** show average mpg for each cylinder group

```
1 library(tidyverse)
2 mtcars |>
3   group_by(cyl) |>
4   summarise(avg_mpg = mean(mpg))
```

A tibble: 3 × 2

	cyl	avg_mpg
	<dbl>	<dbl>
1	4	26.7
2	6	19.7
3	8	15.1

 Try to run the code in your Script Editor.

Data visualization with ggplot2

- Part of the **tidyverse**
- Used for **data visualization**
- Follows the **Grammar of Graphics**: layers of data, aesthetics, and geometries
- The **ggplot** function has three main components: **data**, aesthetics (**aes**), and geometries (**geoms**).

Scatter plot example:

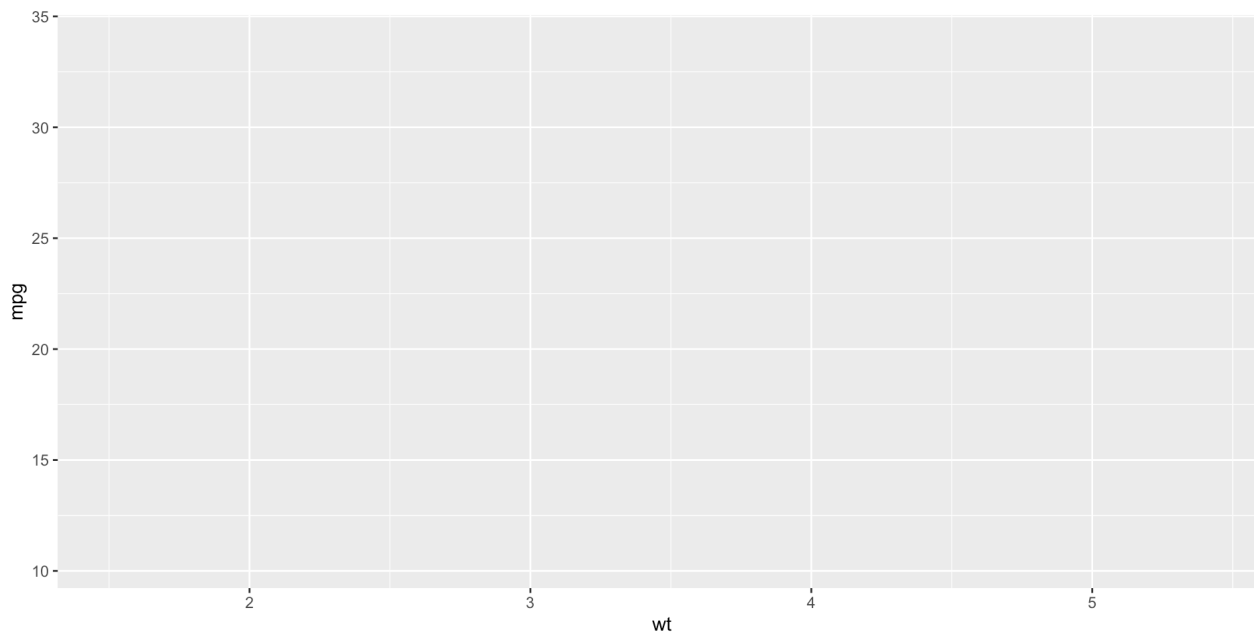
```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = hp, y = mpg)) +
3   geom_point()
```

 Try to run the code in your Script Editor.

aes()

- `aes()` stands for **aesthetics**
- Maps data **columns** to **visual properties** like: x, y positions, color, fill, size, shape, and alpha
- **Example:** map `wt` to x-axis, `mpg` to y-axis, and `cyl` to color

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = wt, y = mpg, color = cyl))
```

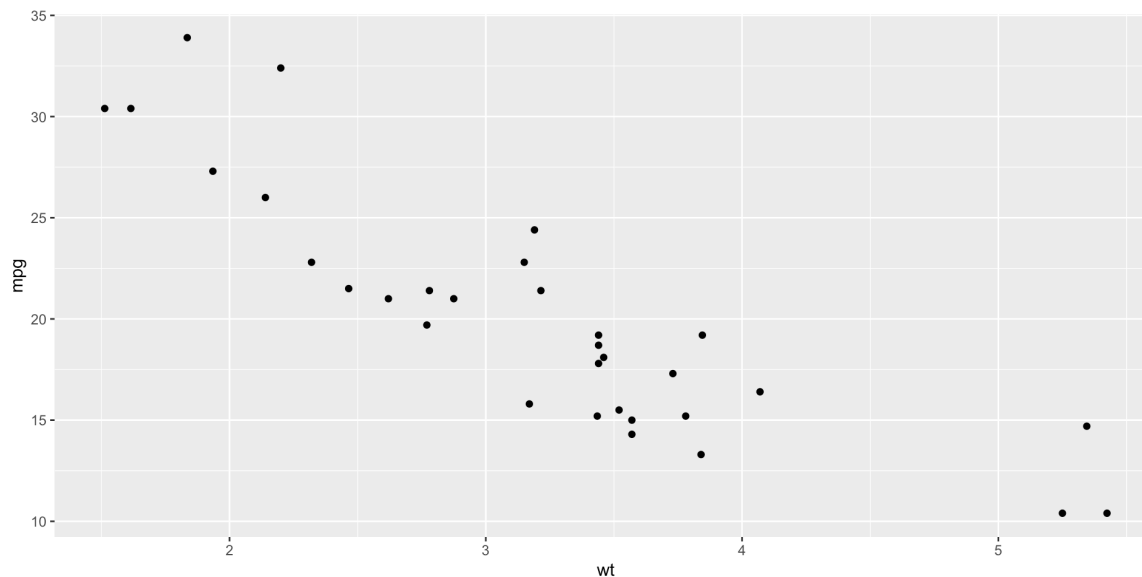


 Try to run the code in your Script Editor.

geom_point()

- **Purpose:** scatter plots
- **Plots points** at x and y positions
- **Example:** visualize the relationship between weight and mpg

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = wt, y = mpg)) +
3   geom_point()
```

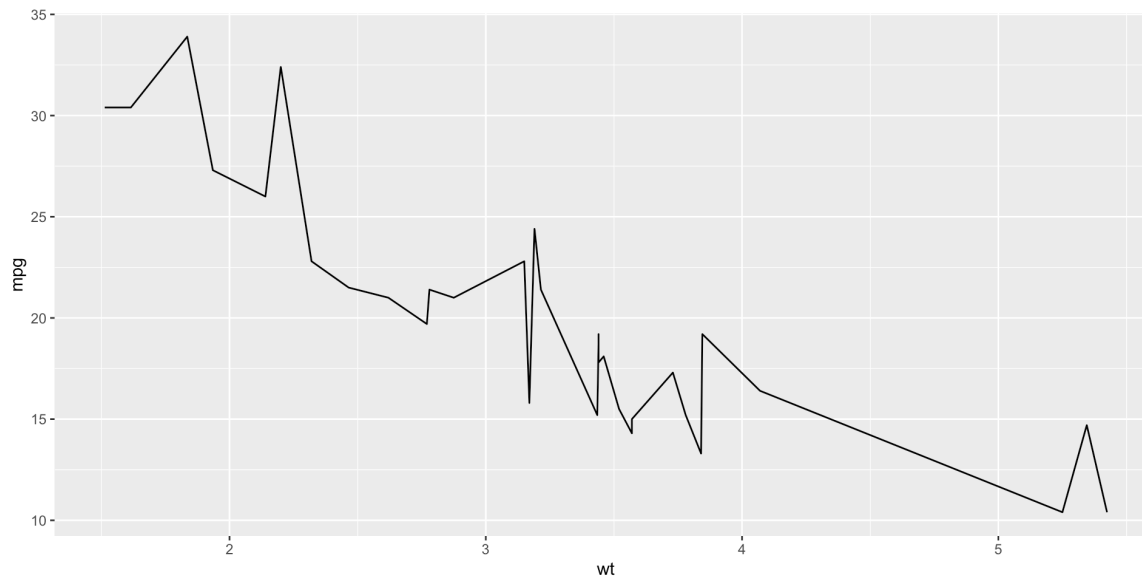


Try to run the code in your Script Editor.

geom_line()

- **Purpose:** line plots
- Connects data points with a **line**, often used for time series
- **Example:** show how unemployment changes over time

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = wt, y = mpg)) +
3   geom_line()
```

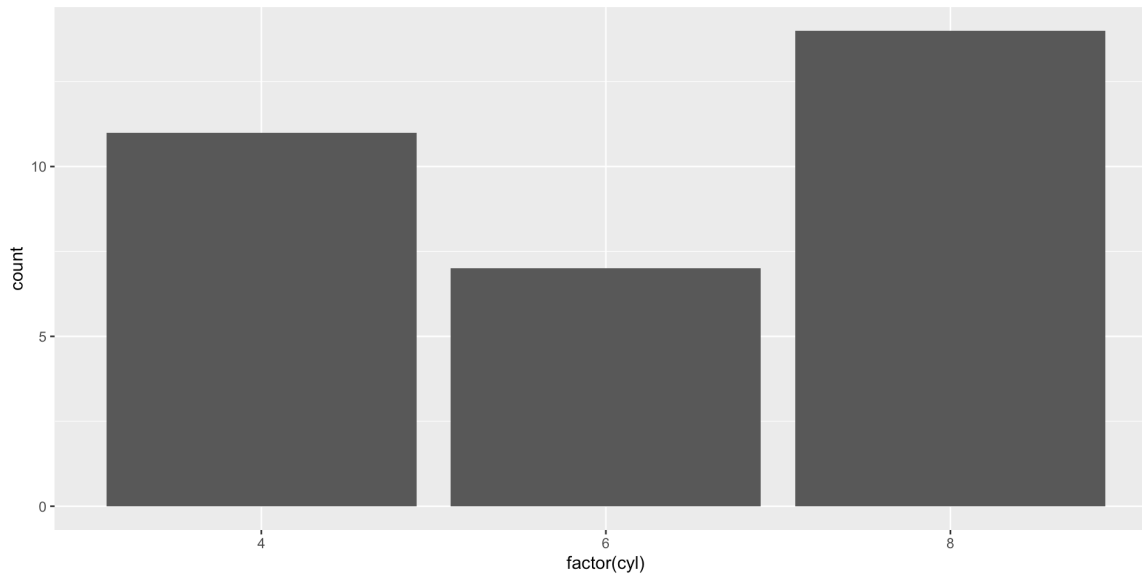


Try to run the code in your Script Editor.

geom_bar()

- **Purpose:** bar charts
- Visualizes **counts or summaries** of categorical data
- **Example:** counts the number of cars for each cylinder group

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = factor(cyl))) +
3   geom_bar()
```

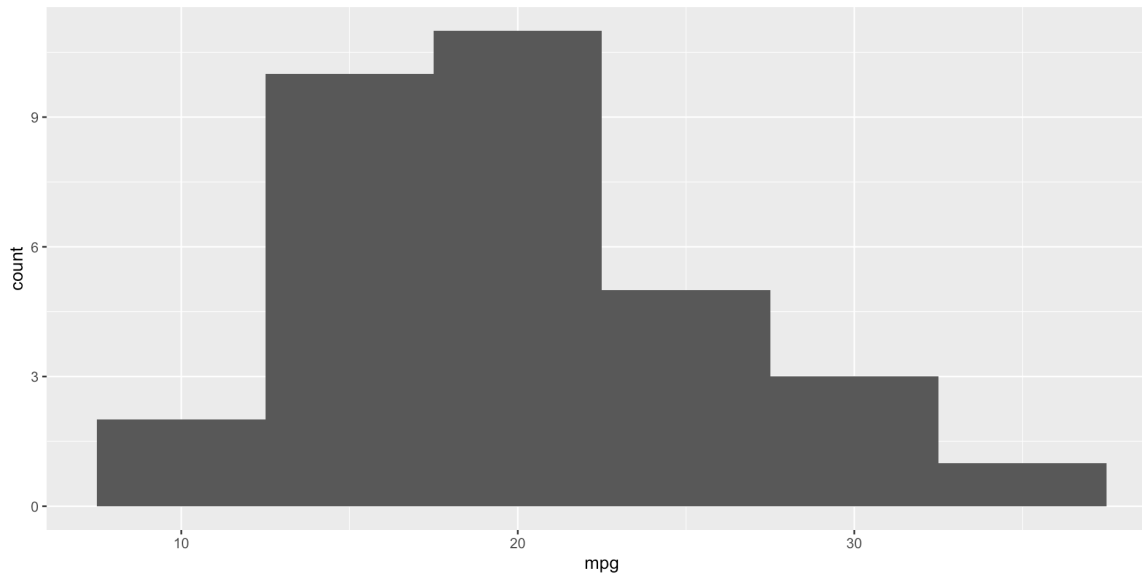


 Try to run the code in your Script Editor.

geom_histogram()

- **Purpose:** histogram for continuous data
- Groups data into bins to show **distribution**
- **Example:** show the distribution of mpg values

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = mpg)) +
3   geom_histogram(binwidth = 5)
```

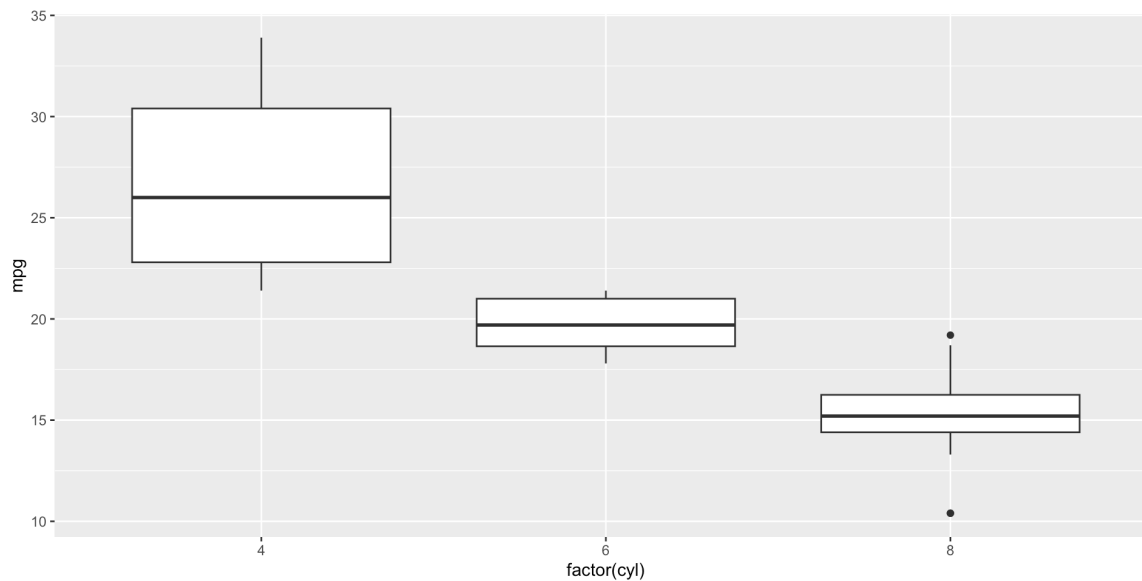


 Try to run the code in your Script Editor.

geom_boxplot()

- **Purpose:** boxplots for distribution and outliers
- Summarizes **median, quartiles, and extremes**
- **Example:** compare mpg across cylinder groups

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +
3   geom_boxplot()
```

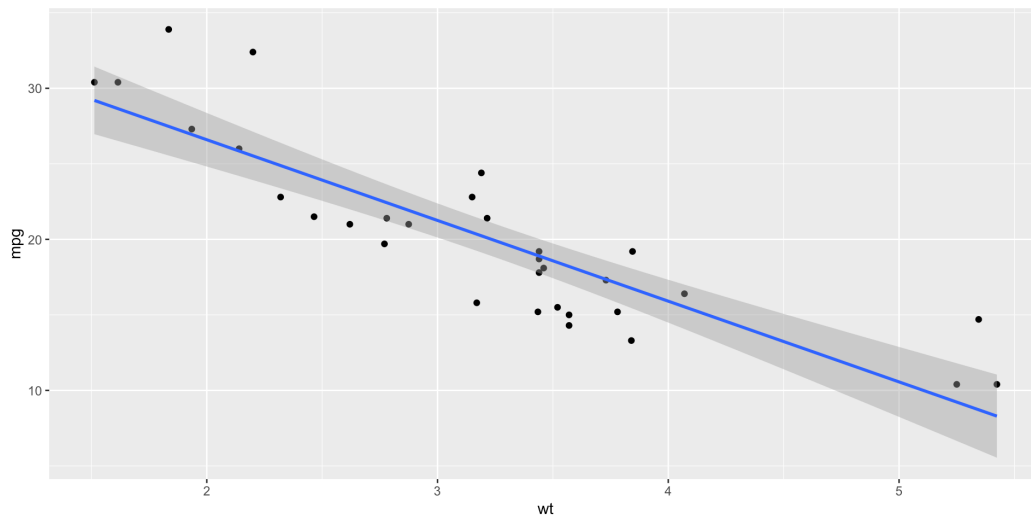


Try to run the code in your Script Editor.

geom_smooth()

- **Purpose:** add trend lines / regression lines
- Can show **linear models** or **smoothed trends**
- **Example:** add a linear regression line to the scatter plots

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = wt, y = mpg)) +
3   geom_point() +
4   geom_smooth(method = "lm")
```



 Try to run the code in your Script Editor.

Data cleaning with `tidyr`

- Part of the **tidyverse**
- Focused on **reshaping and tidying data**
- Make datasets **long and tidy** so that each variable is a column, each observation a row, and each value a cell

Function	Purpose	Example
<code>pivot_longer()</code>	Converts wide data to long (columns → rows)	<code>pivot_longer(df, cols = c(a, b), names_to="var", values_to="val")</code>
<code>pivot_wider()</code>	Converts long data to wide (rows → columns)	<code>pivot_wider(df, names_from=var, values_from=val)</code>
<code>separate()</code>	Splits one column into multiple columns	<code>separate(df, col, into=c("x","y"), sep="-")</code>

Function	Purpose	Example
<code>unite()</code>	Combines multiple columns into one	<code>unite(df, "date", year, month, sep="-")</code>
<code>drop_na()</code>	Removes rows with missing values	<code>drop_na(df)</code>
<code>fill()</code>	Fills missing values with previous/next values	<code>fill(df, year, .direction="down")</code>
<code>replace_na()</code>	Replaces missing values with a specified value	<code>replace_na(df, list(x=0, y="unknown"))</code>
<code>nest()</code>	Creates nested (list-column) data frames	<code>nest(df, data = c(x, y))</code>
<code>unnest()</code>	Expands nested data frames back to flat format	<code>unnest(df, data)</code>

tidyr::pivot_longer()

- **Purpose:** convert wide data to long data
- **Example:** turn multiple stock columns into key-value pairs

```
1 library(tidyverse)
2 stocks <- data.frame(
3   year = 2015:2016,
4   stockA = c(10, 20),
5   stockB = c(15, 25)
6 )
7
8 stocks |>
9 pivot_longer(cols = c(stockA, stockB), names_to = "stock", values_to = "pri
```

```
# A tibble: 4 × 3
  year stock  price
<int> <chr>   <dbl>
1  2015 stockA     10
2  2015 stockB     15
3  2016 stockA     20
4  2016 stockB     25
```

 Try to run the code in your Script Editor.

tidyr::pivot_wider()

- **Purpose:** convert long data to wide data
- **Example:** spread stock names into separate columns

```
1 library(tidyverse)
2 long_data <- data.frame(
3   year = c(2015, 2015, 2016, 2016),
4   stock = c("A", "B", "A", "B"),
5   price = c(10, 15, 20, 25)
6 )
7
8 long_data |>
9 pivot_wider(names_from = stock, values_from = price)
```

```
# A tibble: 2 × 3
  year      A      B
  <dbl> <dbl> <dbl>
1  2015     10     15
2  2016     20     25
```

 Try to run the code in your Script Editor.

tidyr::separate()

- **Purpose:** split one column into multiple
- **Example:** split "2020-01" into year and month columns

```
1 library(tidyverse)
2 df <- data.frame(date = c("2020-01", "2020-02"))
3
4 df |>
5 separate(date, into = c("year", "month"), sep = "-")
```

	year	month
1	2020	01
2	2020	02

 Try to run the code in your Script Editor.

tidyr::unite()

- **Purpose:** combine multiple columns into one
- **Example:** combine year and month into a single "date" column

```
1 library(tidyverse)
2 df <- data.frame(year = c(2020, 2020), month = c("01", "02"))
3
4 df |>
5 unite("date", year, month, sep = "-")
```

date

```
1 2020-01
2 2020-02
```

 Try to run the code in your Script Editor.

tidyr::drop_na()

- **Purpose:** remove rows with missing values
- **Example:** remove rows where any column is **NA**

```
1 library(tidyverse)
2 df <- data.frame(a = c(1, NA, 3), b = c("x", "y", NA))
3
4 df |> drop_na()
```

```
  a b
1 1 x
```

 Try to run the code in your Script Editor.

tidyr::fill()

- **Purpose:** fill missing values using previous or next values
- **Example:** fill missing years with the last known value

```
1 library(tidyverse)
2 df <- data.frame(
3   year = c(2020, NA, NA, 2021, NA),
4   value = 1:5
5 )
6
7 df |>
8 fill(year, .direction = "down")
```

	year	value
1	2020	1
2	2020	2
3	2020	3
4	2021	4
5	2021	5

 Try to run the code in your Script Editor.

And many more

Take home

- The **tidyverse** provides a unified, consistent toolkit for modern data science in R
- Pipes (`|>` / `%>%`) make your code easier to read and write
- **dplyr** simplifies data manipulation with intuitive verbs
- **ggplot2** enables powerful and flexible data visualization
- **tidyr** helps keep your data clean and tidy for analysis

Resources

Tidyverse Website – <https://www.tidyverse.org>

RStudio Cheat Sheets – <https://posit.co/resources/cheatsheets/>

Data Visualization with ggplot2 – <https://ggplot2.tidyverse.org>

Posit Community Forum – <https://community.rstudio.com>

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly Media. Retrieved from <https://r4ds.hadley.nz>

Exercise

Please download the R script with exercises from **GitHub** and try to complete it.